Topic:

Lorentz Transformation

The Lorentz transformations are a set of mathematical rules used to change measurements (like time and distance) from one person's point of view to another's when they are moving at a constant speed. These rules are the foundation of Albert Einstein's theory of special relativity.

1. The Two Main Rules (Postulates)

To derive these equations, we must start with two basic principles:

The Principle of Relativity: The laws of physics are the same in all "inertial frames" (frames moving at a steady speed in a straight line).

The Invariance of the Speed of Light (c): Light in a vacuum always travels at the same speed (c), no matter how fast the source or the observer is moving.

2. The Setup (Standard Configuration)

To make the math easier, we use a "standard setup":

- Imagine two observers, Frame F (standing still) and Frame F' (moving).
- Frame F' moves at a constant speed v along the x-axis.
- Both observers start their clocks at the exact same moment when their positions overlap ($t=t'=0$ and $x=x'=0$)
- Because the motion is only sideways, the up-and-down (y) and forward-back (z) measurements stay the same ($y'=y$ and $z'=z$)

3. The Logic of the Derivation

The goal is to find equations where the speed of light is the same for both people.

- The Light Pulse: Imagine a flash of light goes off at the start. For the person standing still (Frame F), the light travels a distance $x=ct$.
- The Requirement: For the moving person (Frame F'), the light must also travel at speed c, so their measurement must be $x'= ct'$.
- The Spacetime Interval: This means that the mathematical "interval" ($c2t2-x2$) must be invariant, meaning it equals zero for a light pulse in both frames ($c2t2-x2=c2t'2-x'2=0$)
- Linearity: The transformation must be linear (simple proportional math) to ensure that objects moving in a straight line at a steady speed in one frame do the same in the other.

- The Lorentz Factor (γ): To make the math work so that light speed is constant, we need a scaling factor called gamma (γ). This factor depends on the speed v and the speed of light c:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

4. The Final Equations

By solving these requirements, we get the Lorentz transformation equations:

- Time: $t'=\gamma(t-c2vx)$
- Position (x): $x'=\gamma(x-vt)$
- Position (y): $y'=y$
- Position (z): $z'=z$

5. What This Means for Reality

These equations show that space and time are not separate but are part of a 4-dimensional spacetime. This leads to three famous effects:

- Relativity of Simultaneity: Two events that happen at the same time for one person might happen at different times for someone moving.
- Time Dilation: A moving clock will appear to tick slower than a clock at rest.
- Length Contraction: A moving object will appear shorter in the direction it is traveling.